



2014 Assessment Examination

# FORM VI

## MATHEMATICS EXTENSION 2

Thursday 15th May 2014

### General Instructions

- Writing time — 1 hour 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

### Total — 55 Marks

- All questions may be attempted.

### Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

### Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

### Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eight.
- Write your candidate number on this question paper and submit it with your answers.

### Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 79 boys

Examiner  
RCF

**SECTION I - Multiple Choice**

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

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**QUESTION ONE**

The remainder when  $P(z) = z^3 - 3z^2 + 5z - i$  is divided by  $(z - 2i)$  is:

1

- (A)  $12 - 3i$       (B)  $12 + i$       (C)  $6 - 3i$       (D)  $6 + i$

**QUESTION TWO**

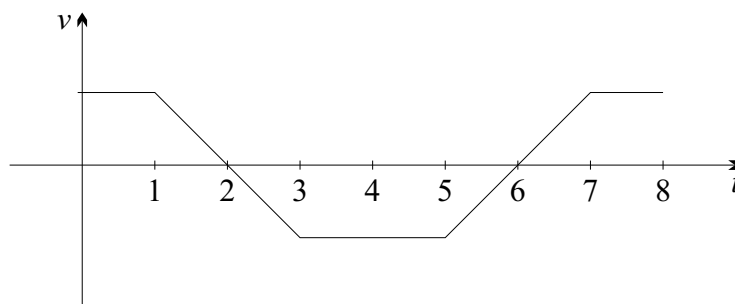
A hyperbola is defined parametrically by the equations  $x = 3\sec\theta$  and  $y = \tan\theta$ .  
Its eccentricity is:

1

- (A)  $\frac{2\sqrt{2}}{3}$       (B) 3      (C)  $\frac{\sqrt{10}}{3}$       (D)  $\sqrt{3}$

**QUESTION THREE**

1



An object moves horizontally with velocity  $v$  at time  $t$  seconds as shown on the graph above. At which time does the object first return to its initial position?

- (A) 2 s      (B) 4 s      (C) 6 s      (D) 8 s

**QUESTION FOUR**

A particle is moving in simple harmonic motion along a straight line such that

1

$$v^2 = 4(2 + 10x - x^2)$$

where  $v$  is the velocity and  $x$  is the displacement. The period  $T$  and centre  $x_0$  are:

- (A)  $T = \pi$  and  $x_0 = 5$       (B)  $T = 2\pi$  and  $x_0 = -5$   
(C)  $T = \pi$  and  $x_0 = -5$       (D)  $T = 2\pi$  and  $x_0 = 5$

**Exam continues next page ...**

**QUESTION FIVE**

Given the hyperbola  $\frac{3x^2}{2} - \frac{2y^2}{3} = 1$ , the gradient of the normal at the point  $(1, \frac{\sqrt{3}}{2})$  is:

**1**

(A)  $\frac{3\sqrt{3}}{2}$

(B)  $-2\sqrt{3}$

(C)  $-\frac{2\sqrt{3}}{9}$

(D)  $-\frac{3\sqrt{3}}{2}$

**QUESTION SIX**

A cannonball is fired on level ground with a given initial velocity. During its subsequent motion it behaves as a projectile accelerating under gravity but experiencing negligible air resistance.

**1**

Which of the following statements about its subsequent motion is INCORRECT?

- (A) At its maximum height the cannonball has velocity equal to the horizontal component of its initial velocity.
- (B) The same range would be achieved using the complementary angle of projection.
- (C) When it lands, the cannonball hits the ground with the same speed as it initially left the cannon.
- (D) At its maximum height the cannonball experiences zero acceleration.

**QUESTION SEVEN**

It is known that the three roots of the cubic equation  $x^3 + 2x^2 + 4x + 8 = 0$  form a geometric progression. The second term in this geometric progression is:

**1**

(A) 2

(B)  $-2i$

(C)  $-2$

(D)  $2i$

\_\_\_\_\_ End of Section I \_\_\_\_\_

Exam continues overleaf ...

## SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

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**QUESTION EIGHT** (12 marks) Use a separate writing booklet.

**Marks**

(a) The equation  $z^3 - 7z^2 + 17z - 15 = 0$  has a root  $2 + i$ .

(i) Explain why  $2 - i$  is also a root of the equation.

**1**

(ii) Find the third root of the equation.

**1**

(b) The ellipse  $\mathcal{E}$  has equation  $3x^2 + 4y^2 = 12$ .

(i) Show that the eccentricity of  $\mathcal{E}$  is  $\frac{1}{2}$ .

**1**

(ii) Find the coordinates of the foci  $S$  and  $S'$ .

**1**

(iii) Find the equations of the two directrices.

**1**

(iv) Sketch  $\mathcal{E}$ , clearly showing its  $x$  and  $y$  intercepts, foci and directrices.

**2**

(v) Find the equation of the tangent to  $\mathcal{E}$  at the point  $(1, \frac{3}{2})$ .

**2**

(c) Find the values of  $k$  for which the polynomial  $P(x) = x^3 + 4x^2 - 3x + k$  has a double zero.

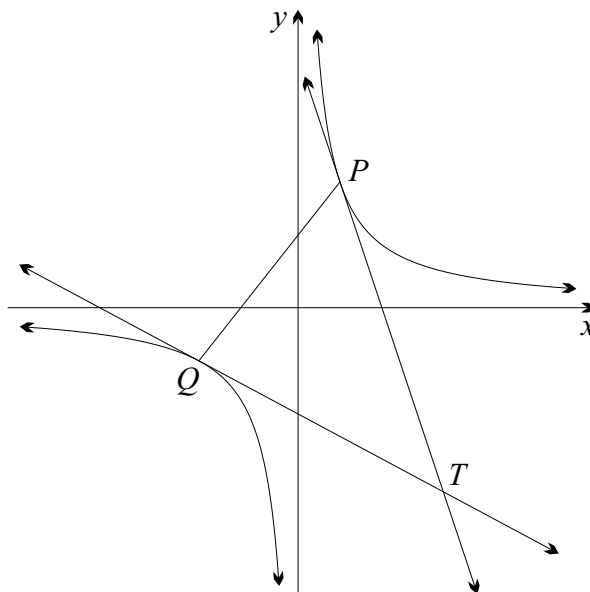
**3**

**QUESTION NINE** (12 marks) Use a separate writing booklet.**Marks**

- (a) Suppose that  $P(x) = x^3 + ax^2 + bx + 8$ , where  $a$  and  $b$  are real.
- (i) Show  $P(4i) = (8 - 16a) + (4b - 64)i$  1
- (ii) When  $P(x)$  is divided by  $x^2 + 16$ , the remainder is  $40 - 11x$ .  
Find the values of  $a$  and  $b$ . 2
- (b) (i) The equation  $x^3 - 4x + 6 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ . Find a polynomial equation with integer co-efficients that has roots  $\alpha^2$ ,  $\beta^2$  and  $\gamma^2$ . 2
- (ii) Hence, or otherwise, find the value of  $\alpha^{-2} + \beta^{-2} + \gamma^{-2}$ . 2
- (c) A truck of mass 2000 kilograms driving along a straight road is propelled by a constant driving force from the engine of 1250 Newtons and experiences a resistive force of magnitude  $2v^2$  where  $v$  metres per second is the truck's velocity. The truck started from rest.
- (i) Show that  $\ddot{x} = \frac{625 - v^2}{1000}$ . 1
- (ii) The truck approaches a maximum speed  $U$ . Find the value of  $U$ , giving your answer in metres per second. 1
- (iii) Find the distance it has travelled when it reaches a speed of  $\frac{U}{2}$ . Give your answer correct to the nearest metre. 3

**QUESTION TEN** (12 marks) Use a separate writing booklet.

**Marks**



- (a) The points  $P(cp, \frac{c}{p})$  and  $Q(cq, \frac{c}{q})$  are on opposite branches of the rectangular hyperbola  $xy = c^2$ , where  $c > 0$ .

(i) Show that the equation of the chord  $PQ$  is  $x + pqy = c(p + q)$ . **2**

(ii) The equation of the tangent at  $P$  is  $x + p^2y = 2cp$ . Show that the tangents at  $P$  and  $Q$  intersect at the point  $T\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ . **2**

(iii) The chord  $PQ$  passes through the fixed point  $(0, c)$ . Describe the locus of  $T$ , including any restrictions. **3**

- (b) Let  $P(z) = 3z^8 - 10z^4 + 3$  and suppose that  $\omega$  is a root of  $P(z) = 0$ .

(i) Show that  $i\omega$  and  $\frac{1}{\omega}$  are also roots of  $P(z) = 0$ . **2**

(ii) Show that  $z = \sqrt[4]{3}$  is a root of  $P(z) = 0$ . **1**

(iii) Hence find all the roots of  $P(z) = 0$ . **2**

**QUESTION ELEVEN** (12 marks) Use a separate writing booklet.**Marks**

- (a) A ball of mass 4 kilograms is thrown vertically upwards from the ground with an initial speed of 12 metres per second. The ball is subject to a downwards gravitational force of magnitude 40 Newtons and air resistance of  $\frac{v^2}{5}$  Newtons in the opposite direction to the velocity  $v$  metres per second.

- (i) If we take the origin at ground level and upwards as the positive direction then the acceleration, until the ball reaches its highest point, is given by

$$\ddot{y} = -10 - \frac{v^2}{20}$$

where  $y$  metres is its height. (Do NOT prove this.)

- ( $\alpha$ ) Use the fact that  $\ddot{y} = v \frac{dv}{dy}$ , to show that, while the ball is rising,

2

$$v^2 = 344e^{-\frac{y}{10}} - 200.$$

- ( $\beta$ ) Hence find the maximum height reached in exact form.

1

- (ii) For the subsequent downwards motion, take downwards as the positive direction and define a new origin at the maximum height.

- ( $\alpha$ ) Find the acceleration for the downwards motion.

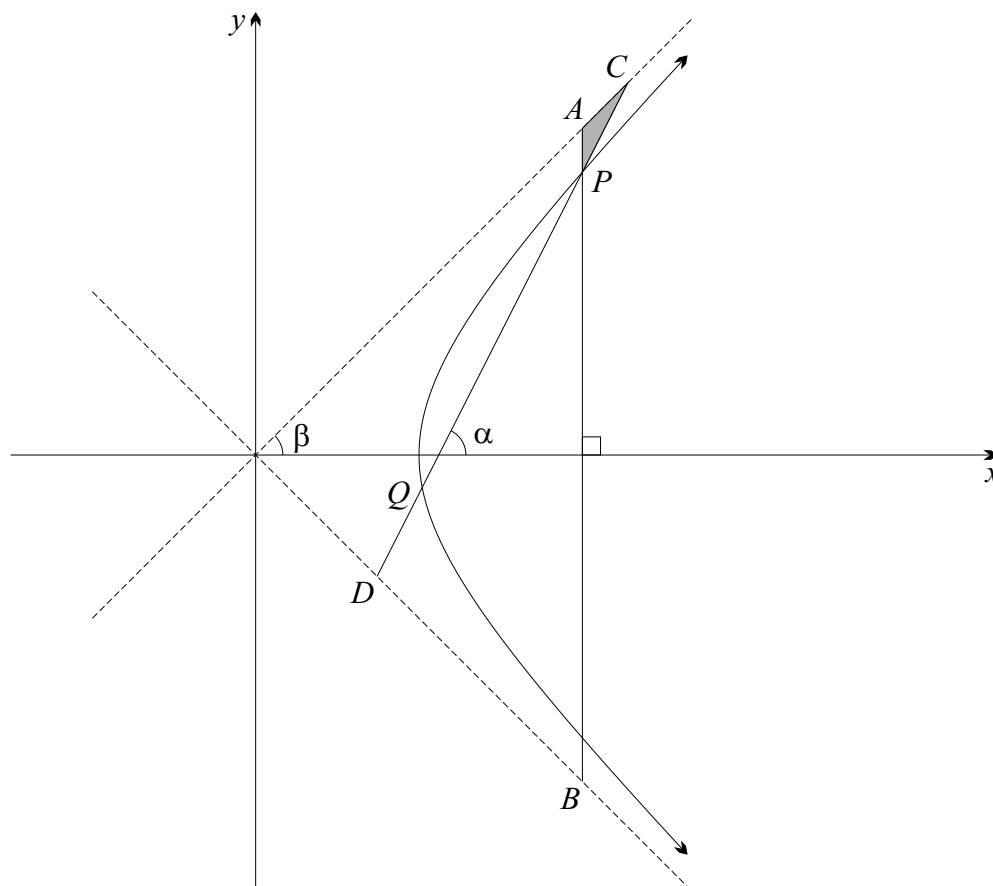
1

- ( $\beta$ ) Hence find the speed at which the ball returns to the ground. Give your answer in metres per second, correct to two decimal places.

2

**QUESTION ELEVEN** (Continued)

(b)



The point  $P(a \sec \theta, b \tan \theta)$  is a point on the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ .

The line through  $P$  perpendicular to the  $x$ -axis meets the asymptotes of the hyperbola  $y = \frac{bx}{a}$  and  $y = -\frac{bx}{a}$  at  $A(a \sec \theta, b \sec \theta)$  and  $B(a \sec \theta, -b \sec \theta)$  respectively.

A second line through  $P$ , with gradient  $\tan \alpha$ , meets the hyperbola again at  $Q$  and meets the asymptotes at  $C$  and  $D$  as shown. The asymptote  $y = \frac{bx}{a}$  makes a fixed angle  $\beta$  with the positive  $x$ -axis at the origin.

(i) Show that  $AP \times PB = b^2$ .

1

(ii) Use the sine rule in  $\triangle ACP$  to show that  $CP = \frac{AP \cos \beta}{\sin(\alpha - \beta)}$ .

3

Also show that  $PD = \frac{PB \cos \beta}{\sin(\alpha + \beta)}$ .

(iii) It follows from part (ii) that  $CP \times PD$  is independent of  $\theta$ . Consequently its value does not depend upon the choice of  $P$ . Hence deduce that  $CP = QD$ .

2

————— End of Section II —————

**END OF EXAMINATION**

B L A N K   P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER: .....

**Question One**

A ☐ B ☐ C ☐ D ☐

**Question Two**

A ☐ B ☐ C ☐ D ☐

**Question Three**

A ☐ B ☐ C ☐ D ☐

**Question Four**

A ☐ B ☐ C ☐ D ☐

**Question Five**

A ☐ B ☐ C ☐ D ☐

**Question Six**

A ☐ B ☐ C ☐ D ☐

**Question Seven**

A ☐ B ☐ C ☐ D ☐

MULTI-CHOICE (1 MARK EACH)

# 4U MAY ASSESSMENT 2014

①  $P(z) = z^3 - 3z^2 + 5z - i$

$$P(2i) = (2i)^3 - 3(2i)^2 + 5(2i) - i$$

$$= -8i + 12 + 10i - i$$

$$= 12 + i \quad (B)$$

②  $x = 3 \sec \theta$   $y = \tan \theta$

$$\frac{x^2}{3^2} - y^2 = 1$$

$$\therefore a=3, b=1$$

$$b^2 = a^2(e^2 - 1)$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$e^2 = 1 + \frac{1}{9}$$

$$e = \frac{\sqrt{10}}{3}$$

(C)

③ Area under velocity time graph represents displacement hence back to starting point when area above and below axis equal ie first @ 15 (B)

④  $v^2 = 4(2+10x-x^2)$

$$\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2) = \frac{d}{dx}(4+20x-2x^2)$$

$$= 20 - 4x$$

$$= -4(x-5)$$

$$\therefore n^2 = 4$$

$$n = 2$$

$$T = \frac{2\pi}{n} = \pi/5$$

(A)

⑤  $\frac{3x^2 - 2y^2}{2} = 1$

$$\text{Diff w.r.t } x \quad 3x - \frac{4y}{3} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{9x}{4y}$$

Grad of normal is  $-\frac{4y}{9x}$

$$m = -\frac{4(\frac{\sqrt{3}}{2})}{9} = -\frac{2\sqrt{3}}{9} \quad (C)$$

⑥ Only incorrect statements is (D). Cannonball experiences constant acceleration due to gravity for entire motion.

⑦  $x^3 + 2x^2 + 4x + 8 = 0$

Roots in G.P.

Let roots be  $\frac{a}{r}, a, ar$

$$\sum \alpha\beta\gamma = -\frac{8}{1}$$

$$\therefore a^3 = -8$$

$$\therefore a = (-2)$$

Second term is  $(-2)$

(C)

⑧ a)  $z^3 - 7z^2 + 17z - 15 = 0$

(i) Since the equation has REAL CO-EFFICIENTS any complex roots appear in CONJUGATE PAIRS. ✓

(ii)  $\sum \alpha = (2+i)(2-i) + \alpha$

$$= 4 + \alpha$$

$$= -(-7)$$

$$\therefore \alpha = 3 \quad \checkmark$$

OR  $\sum \alpha\beta\gamma = -(-15)$

$$\therefore (2+i)(2-i)\alpha = 15$$

$$5\alpha = 15$$

$$\alpha = 3.$$

b)  $3x^2 + 4y^2 = 12$

(i) ( $\div 12$ )

$$\frac{x^2}{4} + \frac{y^2}{3} = 1.$$

$$a^2 = 4 \quad a = 2$$

$$b^2 = 3 \quad b = \sqrt{3}.$$

$$b^2 = a^2(1 - e^2)$$

$$\therefore e^2 = 1 - \frac{b^2}{a^2}$$

✓ SHOW

$$e^2 = 1 - \frac{3}{4}$$

$$e = \frac{1}{2} \quad (e > 0)$$

(ii) For  $S$  &  $S'$  are  $(ae, 0)$  and  $(-ae, 0)$  ✓  
ie  $(1, 0)$  and  $(-1, 0)$  ✓

(iii) Directrices  $x = \pm \frac{a}{e}$   
 $= \pm \frac{2}{1/2}$   
 $x = \pm 4$  ✓

(iv)  $3x^2 + 4y^2 = 12$

$$\text{Diff w.r.t } x \quad 6x + 8y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{6x}{8y}$$

$$\therefore m = -\frac{6}{8 \times \frac{3}{2}}$$

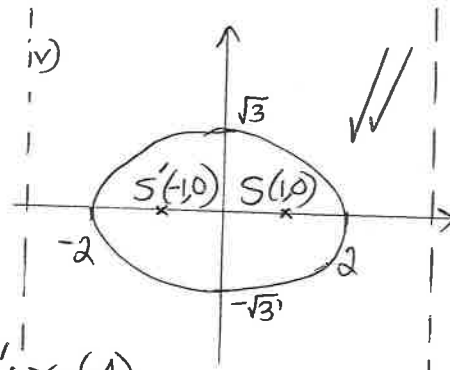
$$= -\frac{1}{2} \quad \checkmark$$

Eqn of tangent

$$y - \frac{3}{2} = -\frac{1}{2}(x - 1)$$

$$2y - 3 = 1 - x$$

$$x + 2y - 4 = 0. \quad \checkmark$$



$$a) P(x) = x^3 + 4x^2 - 3x + k$$

$$P'(x) = 3x^2 + 8x - 3$$

$$= (3x-1)(x+3)$$

$$P'(x) = 0 \Rightarrow x = \frac{1}{3} \text{ or } (-3)$$

$$P\left(\frac{1}{3}\right) = \left(\frac{1}{3}\right)^3 + 4\left(\frac{1}{3}\right)^2 - 3\left(\frac{1}{3}\right) + k$$

$$= \frac{1}{27} + \frac{4}{9} - 1 + k$$

$$= \frac{27k + 1 + 12 - 27}{27}$$

$$= \frac{27k - 14}{27}$$

$$P\left(\frac{1}{3}\right) = 0 \quad k = \frac{14}{27}$$

$$P(-3) = (-3)^3 + 4(-3)^2 - 3(-3) + k$$

$$= -27 + 36 + 9 + k$$

$$= 18 + k$$

$$\text{If } P(-3) = 0, \quad k = (-18)$$

$$9) P(x) = x^3 + ax^2 + bx + 8$$

$$a) (i) P(4i) = (4i)^3 + a(4i)^2 + b(4i) + 8$$

$$= 64i^3 + 16i^2a + 4bi + 8$$

$$= -64i - 16a + 4bi + 8$$

$$= (8 - 16a) + (4b - 64)i \quad \text{①}$$

$$(ii) P(x) = (x - 4i)(x + 4i)Q(x) + 40 - 11x$$

$$(x^2 + 16)Q(x) + 40 - 11x$$

$$\therefore P(4i) = 40 - 11 \times 4i \quad \text{②}$$

$$\therefore \text{equating real parts of ① + ②}$$

$$40 = 8 - 16a$$

$$16a = -32$$

$$a = -2$$

$$\text{equating imag parts}$$

$$4b - 64 = -44$$

$$4b = 20$$

$$b = 5$$

Alternative approach using long division algorithm

$$\begin{array}{r} x + a \\ x^2 + 16 \overline{) x^3 + ax^2 + bx + 8} \\ \underline{x^3 \phantom{+ 16x} -} \\ ax^2 + (b-16)x + 8 \\ \underline{ax^2 + 0x + 16a -} \\ (b-16)x + 8 - 16a \end{array}$$

equating:

coef of x

$$(b-16) = -11$$

$$\Rightarrow b = 5$$

constants

$$(8 - 16a) = 40$$

$$\Rightarrow a = -2$$

b) (i)  $P(x) = x^3 - 4x + 6 = 0$  has roots  $\alpha, \beta, \gamma$   
 Eqn with roots  $\alpha^2, \beta^2, \gamma^2$  requires replacing  $x$  with  $\sqrt{x}$

$$P(\sqrt{x}) = (\sqrt{x})^3 - 4(\sqrt{x}) + 6 = 0$$

$$= \sqrt{x}(x-4) + 6 = 0$$

$$\therefore \sqrt{x}(x-4) + 6 = 0$$

$$\sqrt{x}(x-4) = -6$$

$$(sq) \quad (sq)$$

$$x(x-4) = 36$$

$$x(x^2 - 8x + 16) = 36$$

$$x^3 - 8x^2 + 16x - 36 = 0$$

has roots  $\alpha^2, \beta^2, \gamma^2$

$$(ii) \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{\sum \alpha\beta}{\sum \alpha\beta\gamma}$$

$$= \frac{16}{-36}$$

$$= -\frac{4}{9}$$

Alternative Approaches forming NEW eqns with roots  $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$

a) Use  $x^3 - 8x^2 + 16x - 36 = 0$  and replace  $x$  with  $\frac{1}{x}$

$$\frac{1}{x^3} - \frac{8}{x^2} + \frac{16}{x} - 36 = 0$$

$$(\times x^3)$$

$$1 - 8x + 16x^2 - 36x^3 = 0$$

$$36x^3 - 16x^2 + 8x - 1 = 0$$

$$\text{then (iii) } \sum \alpha = -\frac{b}{a} = \frac{16}{36} = \frac{4}{9}$$

b) Use original  $x^3 - 4x + 6 = 0$  and replace  $x$  with  $\frac{1}{\sqrt{x}}$

$$\left(\frac{1}{\sqrt{x}}\right)^3 - 4\left(\frac{1}{\sqrt{x}}\right) + 6 = 0$$

$$\times x^{\frac{3}{2}}$$

$$1 - 4x + 6x^{\frac{3}{2}} = 0$$

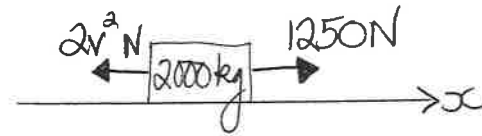
$$6x^{\frac{3}{2}} = 4x - 1$$

$$(sq) \quad (sq)$$

$$36x^3 = 16x^2 - 8x + 1$$

$$36x^3 - 16x^2 + 8x - 1 = 0 \text{ as before}$$

c)



(i) Eqn of motion  $\rightarrow$

$$1250 - 2v^2 = 2000\ddot{x}$$

$$\ddot{x} = \frac{2(625 - v^2)}{2000}$$

$$\ddot{x} = \frac{625 - v^2}{1000}$$

✓ SHAW

(ii) Max Speed  $\ddot{x} = 0$

$$\therefore 625 - v^2 = 0$$

$$v^2 = 625$$

$$v = 25 \text{ m/s } (v > 0)$$

(iii)  $v dv = \frac{625 - v^2}{1000}$

$$\int_0^D \frac{2v}{625 - v^2} dv = \int_0^D \frac{1}{500} dx$$

Initially  $v = 0$   $x = 0$   
 when  $v = \frac{v}{2}$  let  $x = D$ .

$$\left[ -\ln(625 - v^2) \right]_0^D = \left[ \frac{x}{500} \right]_0^D$$

$$-\ln\left(625 - \frac{625}{4}\right) - (-\ln 625) = \frac{D}{500}$$

$$\ln\left(\frac{625}{625(\frac{3}{4})}\right) = \frac{D}{500}$$

$$500 \ln\left(\frac{4}{3}\right) = D$$

Distance travelled is approx. 144 metres ✓

Alternative approach w/o reciprocals and indefinite integration

$$\frac{dv}{dx} = \frac{625-v^2}{1000v}$$

$$\frac{dx}{dv} = \frac{1000v}{625-v^2}$$

$$\int dx \quad \int dv$$

$$x = -500 \ln(625-v^2) + C$$

$$v=0 \quad x=0 \quad 0 = -500 \ln 625 + C$$

$$C = 500 \ln 625$$

$$x = 500 \ln \left( \frac{625}{625-v^2} \right)$$

$$V = \frac{V_0}{2} = \frac{25}{2}$$

$$= 500 \ln \left( \frac{625}{625 \times \frac{1}{3/4}} \right)$$

$$= 500 \ln 3$$

$$\approx 144 \text{ m}$$

⑩

$$\begin{aligned} \text{a) } \omega m_{pq} &= \frac{c/q - c/p}{cq - cp} \\ &= \frac{c(p-q)}{c(pq)} \\ &= -\frac{1}{pq} \quad (p \neq q) \end{aligned}$$

$$\begin{aligned} \therefore \text{Eqn of chord } y - \frac{c}{p} &= -\frac{1}{pq}(x - cp) \quad \checkmark \text{ SHOW} \\ pqy - cq &= -x + cp \\ x + pqy &= c(p+q) \end{aligned}$$

$$\begin{aligned} \text{(ii) Tangent at P} \quad x + p^2 y &= 2cp \quad \text{①} \\ \text{at Q} \quad x + q^2 y &= 2cq \quad \text{②} \end{aligned}$$

Solve simultaneously ①+②

$$2cp - p^2 y = 2cq - q^2 y$$

$$2cp - 2cq = (p^2 - q^2) y$$

$$\frac{2c(p-q)}{(p-q)(p+q)} = y \quad \checkmark \text{ SHOW}$$

$$\therefore y = \frac{2c}{p+q}$$

sub into ①

$$x = 2cp - p^2 \left( \frac{2c}{p+q} \right)$$

$$x = \frac{2cp(p+q) - 2cp^2}{p+q}$$

$$= \frac{2cpq}{p+q} \quad \checkmark \text{ SHOW}$$

$$\therefore T \left( \frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$$

since distinct points on opposite branches of R.

(iii) If chord PQ passes through  $(0, c)$  then  $0 + pqc = c(p+q)$   
 $c \neq 0 \therefore pq = p+q$  ✓  
 $p+q \neq 0$  since  $p \neq 0, q \neq 0$   
 $\therefore pq \neq 0$  ✓

$$T\left(\frac{2c(p+q)}{(p+q)}, \frac{2c}{p+q}\right)$$

$$= T\left(2c, \frac{2c}{pq}\right)$$

also  $pq < 0$  since  
 P and Q on opposite  
 branches of hyperbola

hence locus of T is vertical ray, with eqn  $x = 2c$  ✓  
 where  $y < 0$

✓ Restriction

note: Chord PQ requires a  
 positive gradient for Q to lie on  
 opposite branch hence if P on  
 first quadrant branch  $0 < p < 1$

b)  $P(z) = 3z^8 - 10z^4 + 3$

$P(w) = 0 \therefore 3w^8 - 10w^4 + 3 = 0$

(i)  $P(iw) = 3(iw)^8 - 10(iw)^4 + 3$  ✓  
 $= 3w^8 - 10w^4 + 3$  ✓ (since  $i^8 = i^4 = 1$ )  
 $= 0$

$\therefore iw$  a root of  $P(z) = 0$

$P\left(\frac{1}{w}\right) = 3\left(\frac{1}{w}\right)^8 - 10\left(\frac{1}{w}\right)^4 + 3$   
 $= \frac{3}{w^8} - \frac{10}{w^4} + 3$

$w^8 \times P\left(\frac{1}{w}\right) = 3 - 10w^4 + 3w^8$  ✓  
 $= 0$

but  $w \neq 0$  since  $P(0) = 3$  hence 0 not a soln

$\therefore P\left(\frac{1}{w}\right) = 0 \therefore \frac{1}{w}$  a root of  $P(z) = 0$

(ii)  $P(3^{\frac{1}{4}}) = 3(3^{\frac{1}{4}})^8 - 10(3^{\frac{1}{4}})^4 + 3$   
 $= 3 \times 3^2 - 10 \times 3 + 3$  ✓  
 $= 27 - 30 + 3$   
 $= 0$

$\therefore \sqrt[4]{3}$  is a root of  $P(z) = 0$

since  $P(z)$  has even symmetry  $\bar{z} = -\sqrt[4]{3}$  a root

from (i)  $z = \sqrt[4]{3}i$  also a root, and its conjugate  $-\sqrt[4]{3}i$

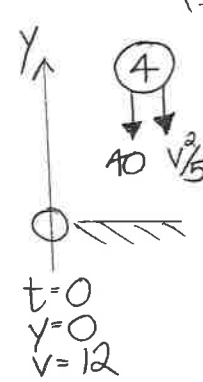
since real coefficients

from (i)  $z = \frac{1}{\sqrt[4]{3}}$  a root, also  $z = -\frac{1}{\sqrt[4]{3}}$

also  $z = \frac{i}{\sqrt[4]{3}}$  and  $z = -\frac{i}{\sqrt[4]{3}}$

Hence eight roots are  $\pm \sqrt[4]{3}, \pm \sqrt[4]{3}i, \pm \frac{1}{\sqrt[4]{3}}, \pm \frac{i}{\sqrt[4]{3}}$

11  
a)



(i) Egn of motion  
 $\uparrow 4\ddot{y} = -10 - \frac{v^2}{5}$   
 $\ddot{y} = -10 - \frac{v^2}{20}$  (Given)

(x)  $v \frac{dv}{dy} = -\left(\frac{200+v^2}{20}\right)$

$\int \frac{2v}{200+v^2} dv = \int -\frac{dy}{10}$

$\ln(200+v^2) = -\frac{y}{10} + C$

when  $y=0$   $v=12$

$\therefore \ln 344 = 0 + C$

$\therefore \ln(200+v^2) - \ln 344 = -\frac{y}{10}$  ✓ show

$\ln \frac{200+v^2}{344} = -\frac{y}{10}$

$\frac{200+v^2}{344} = e^{-y/10}$

$v^2 = 344e^{-y/10} - 200$

(p) Max Height  $v=0$

$\therefore 344e^{-y/10} = 200$

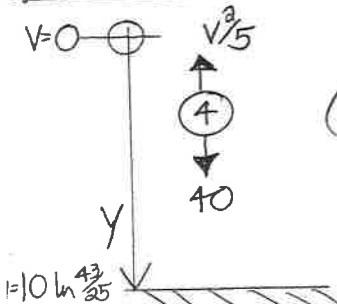
$e^{-y/10} = \frac{200}{344}$

$e^{y/10} = \frac{344}{200}$

$= \frac{43}{25}$

$y = 10 \ln \frac{43}{25}$  metres ✓

(ii) DOWNWARD



Egn of motion ↓  
 (x)  $4\ddot{y} = 40 - \frac{v^2}{5}$   
 $\ddot{y} = 10 - \frac{v^2}{20}$  ✓

(p)

$v \frac{dv}{dy} = \frac{200-v^2}{20}$   
 $\int \frac{-2v}{200-v^2} dv = \int -\frac{dy}{10}$

$\left[ \ln(200-v^2) \right]_0^V = \left[ -\frac{y}{10} \right]_0^{10 \ln \frac{43}{25}}$  ✓

$\ln(200-V^2) - \ln 200 = -\ln \frac{43}{25}$

$\ln(200-V^2) = \ln 200 - \ln \frac{43}{25}$

$= \ln \left( \frac{5000}{43} \right)$

$\therefore 200-V^2 = \frac{5000}{43}$

$V^2 = \frac{8600}{43} - \frac{5000}{43}$

$= \frac{3600}{43}$

hence  $V = \frac{60}{\sqrt{43}}$  m/s. ✓ ( $V > 0$ )

$\approx 9.15$  m/s

$$\begin{aligned} b) \quad AP &= Y_A - Y_P \\ &= b \sec \theta - b \tan \theta \\ &= b(\sec \theta - \tan \theta) \end{aligned}$$

$$\begin{aligned} PB &= Y_P - Y_B \\ &= b \tan \theta - (-b \sec \theta) \\ &= b \tan \theta + b \sec \theta \end{aligned}$$

$$\therefore AP \times PB = b(\sec \theta - \tan \theta) \times b(\tan \theta + \sec \theta) \quad \checkmark \text{ SHAW}$$

$$= b^2 (\sec^2 \theta - \tan^2 \theta)$$

(i) Consider  $\triangle ACP$  (Exterior  $\angle$   
 $\angle ACP = \alpha - \beta$ .  $\triangle CAP$ )  
 $\angle CAP = \beta + \frac{\alpha}{2}$  (Adjacent Angles)

Using sine rule

$$\frac{AP}{\sin(\angle ACP)} = \frac{CP}{\sin(\angle CAP)}$$

$$\therefore \frac{AP}{\sin(\alpha - \beta)} = \frac{CP}{\sin(\beta + \frac{\alpha}{2})}$$

$$\frac{AP}{\sin(\alpha\beta)} = \frac{CP}{\sin \beta} \quad \checkmark \text{ show}$$

$$\therefore OP = \frac{AP \cos \beta}{\sin(\alpha - \beta)}$$

Similarly in  $\triangle PDB$

$\angle PDB = \alpha + \beta$ . (Adjacent  $\angle$ )  
 $\angle PBD = \frac{\pi}{2} - \beta$ . ( $\triangle PBD$ )

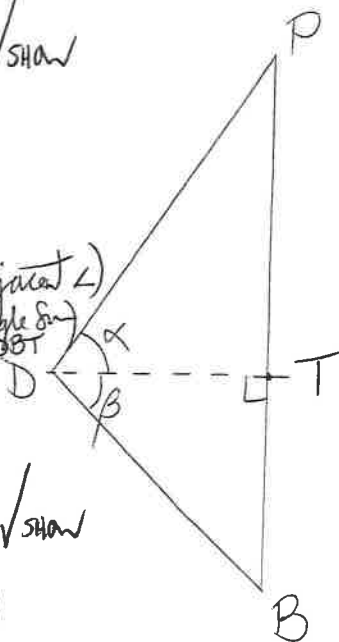
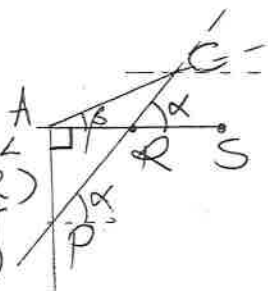
$$\angle PBD = \frac{\pi}{2} - \beta. \quad (\text{Angle in } \triangle DBT)$$

Using sine rule

$$\frac{PD}{\sin(\angle PBD)} = \frac{PB}{\sin(\angle PDB)}$$

$$\frac{PD}{\sin(\frac{\pi}{2}-\beta)} = \frac{PB}{\sin(\alpha+\beta)}$$

$$PD = \frac{PB \cos \beta}{\sin(\alpha + \beta)}$$



$$\begin{aligned} \text{(iii)} \quad CP \times PD &= \frac{AP \cos \beta}{\sin(\alpha - \beta)} \times \frac{PB \cos \beta}{\sin(\alpha + \beta)} \\ &= \frac{AP \times PB \cos^2 \beta}{\sin(\alpha - \beta) \sin(\alpha + \beta)} \\ &= \frac{b^2 \cos^2 \beta}{\sin(\alpha - \beta) \sin(\alpha + \beta)} \end{aligned}$$

$b, \beta$  constants  
only variable  $x$

since product doesn't depend upon choice of P

$$CP \times PD = CQ \times QD$$

$$CP \times (PQ + QD) = (CP + PQ) \times QD$$

$$CP \times PQ + \cancel{CP \times QD} = \cancel{CP \times QD} + PQ \times QD$$

$$\therefore CP = QD \quad \blacksquare \quad (\text{since } PQ \neq 0)$$